

CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Lecture Outline

The lecture covers the following topics:

1. **Transformation of Random Variables**
Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
Joint transformations and derived distributions.
3. **Illustrative Example**
Detailed derivation for the case

$$Z = X + Y$$

These topics together explain how the distribution of a **new random variable** can be derived from existing random variables.

1 Transformation of Random Variables

Definition

A **transformation of a random variable** occurs when a new random variable is defined as a function of another random variable.

Let X be a random variable and let

$$Y = g(X)$$

Then Y is called a **transformed random variable**.

The main objective is to determine the **probability distribution of Y** given the probability distribution of X .

2 Distribution of a Transformed Variable

The most common method for finding the distribution of a transformed variable is the **CDF method**.

Step 1: Start with the CDF Definition

The cumulative distribution function of Y is

$$F_Y(y) = P(Y \leq y)$$

Step 2: Substitute the Transformation

If

$$Y = g(X)$$

then

$$F_Y(y) = P(g(X) \leq y)$$

Step 3: Express Probability in Terms of X

The evaluation of this probability depends on whether the function $g(x)$ is:

- Monotonic increasing, or
- Monotonic decreasing

3 Case 1: Monotonic Increasing Transformation

Assume that the function $g(x)$ is **strictly increasing**.

Then

$$g(X) \leq y$$

implies

$$X \leq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

Derivation of the PDF

If the random variable X has probability density function $f_X(x)$, then the PDF of Y is obtained by differentiation.

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substituting the expression of $F_Y(y)$:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the **chain rule**

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus the transformation formula becomes

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

4 Case 2: Monotonic Decreasing Transformation

If $g(x)$ is **strictly decreasing**, then

$$g(X) \leq y$$

implies

$$X \geq g^{-1}(y)$$

Therefore

$$F_Y(y) = P(X \geq g^{-1}(y))$$

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

The probability density function can again be obtained by differentiating this CDF.

5 Function of Two Random Variables

In many situations a new random variable is defined as a **function of two random variables**.

Let

$$Z = h(X, Y)$$

To determine the distribution of Z , we require the **joint probability distribution** of X and Y .

Joint Probability Density Function

If X and Y are continuous random variables, they may have a joint PDF

$$f_{X,Y}(x, y)$$

This function describes the probability behavior of the pair (X, Y) .

Independence Assumption

If X and Y are **independent**, then their joint PDF is

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

This property simplifies the derivation of distributions for functions such as

- $Z = X + Y$
- $Z = X - Y$
- $Z = XY$

6 Illustrative Example

Problem Statement

Find the distribution of

$$Z = X + Y$$

where

- X and Y are **independent random variables**
- Both follow **exponential distributions**

$$X \sim \text{Exp}(\lambda)$$

$$Y \sim \text{Exp}(\lambda)$$

Step 1: Write the PDFs

For an exponential random variable

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

Step 2: Apply the Convolution Formula

For independent variables

$$Z = X + Y$$

the probability density function is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

Step 3: Determine the Limits of Integration

Because exponential variables are defined only for

$$x \geq 0$$

and

$$y = z - x \geq 0$$

we obtain

$$0 \leq x \leq z$$

Therefore

$$f_Z(z) = \int_0^z f_X(x) f_Y(z - x) dx$$

Step 4: Substitute the PDFs

$$f_Z(z) = \int_0^z (\lambda e^{-\lambda x})(\lambda e^{-\lambda(z-x)}) dx$$

Step 5: Simplify the Expression

$$f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx$$

Step 6: Evaluate the Integral

$$\int_0^z dx = z$$

Thus

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

Final Probability Density Function

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Interpretation

The sum of two independent exponential random variables follows a **Gamma (Erlang) distribution** with:

- shape parameter $k = 2$
- rate parameter λ

Key Formulas from Lecture 11

Transformation Formula

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Convolution Formula

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

For non-negative variables:

$$f_Z(z) = \int_0^z f_X(x) f_Y(z-x) dx$$